

TrES-1b

R-2409

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_120620 · created 2026-08-21T01:17:07+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456098.70557 +/- 0.00092 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.114 +/- 0.017
Transit depth (Rp/Rs)^2	1.3 +/- 0.39 [%]
Orbital Inclination (inc)	88.07 +/- 0.89
Airmass coefficient 1 (a1)	1.1 +/- 0.012
Airmass coefficient 2 (a2)	-0.0837 +/- 0.0095
Scatter in the residuals of the lightcurve fit is	1.04 %
Best Comparison Star	None
Optimal Aperture	4.23
Optimal Annulus	14.42
Transit Duration (day)	0.0848 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.114 $\pm$ 0.017 vs 0.1358 (deviation 1.11 σ)
Duration fit vs published	0.0848 d vs 0.1042 d (deviation 5.25 σ)
Mid-time O-C	1.4 min
Depth SNR	5.8
Residual scatter	1.04 %

Light curve

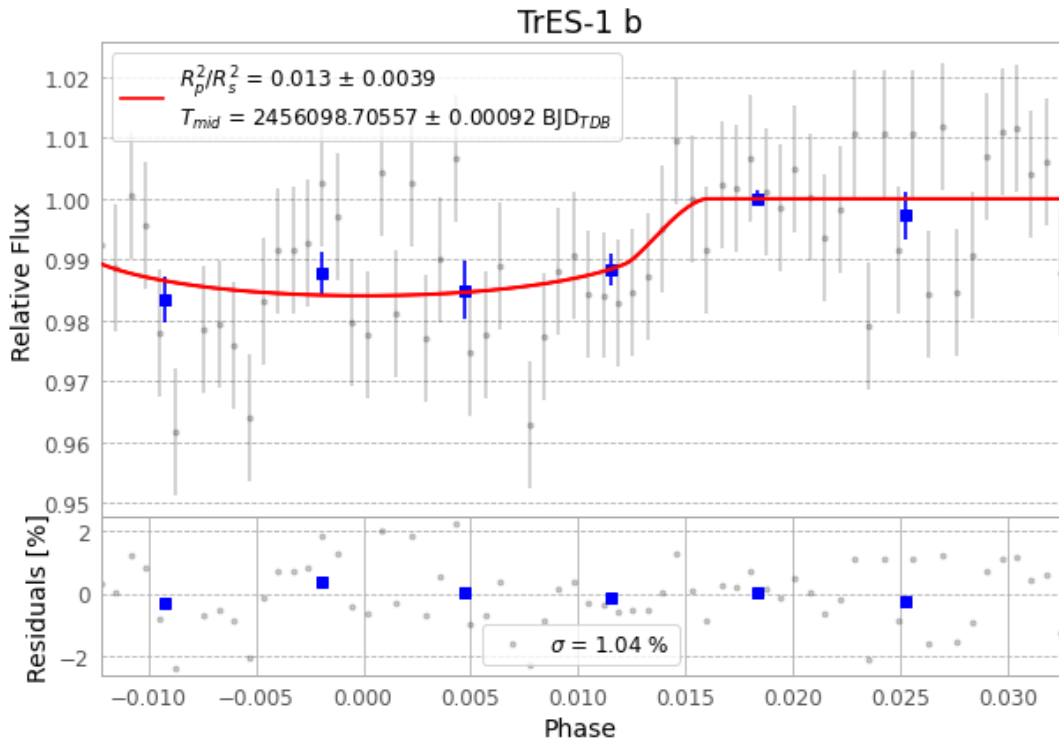


Figure 1 — FinalLightCurve_TrES-1 b_2012-06-19.png (SHA-256 4e9cdaa51cfbb98b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [379, 186], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 20099d4d4a89486d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), TRES-1120620035712.FITS → TRES-1120620071213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-120620.log (a291b1ac7340...), FinalLightCurve_TrES-1 b_2012-06-19.png (4e9cdaa51cfb...), FinalLightCurve_TrES-1 b_2012-06-19.csv (10b00c37b75c...)

Compute resources

Wall time 122.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 01:17Z.